

Development of a metacognitive regulation-based collaborative programming system and its effects on students' learning achievements, computational thinking tendency and group metacognition

Wei Li^{1,2}  | Cheng-Ye Liu¹  | Judy C. R. Tseng³ 

¹STEM Education Research Center, Wenzhou University, Wenzhou, China

²Program in Engineering Science, Chung Hua University, Hsinchu, Taiwan

³Department of Computer Science and Information Engineering, Chung Hua University, Hsinchu, Taiwan

Correspondence

Judy C. R. Tseng, Department of Computer Science and Information Engineering, Chung Hua University, Hsinchu 300, Taiwan.

Email: judyrcr@gmail.com

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Abstract

Collaborative programming helps improve students' computational thinking and increases their confidence in solving programming problems. However, the effect of collaborative learning is not ideal because it is difficult for students to mobilize metacognition to regulate learning spontaneously. To guide students to effectively regulate the learning process when they collaborate to solve programming problems, this study develops a collaborative learning approach and a Collaborative programming System (MR-CPS) based on metacognitive regulation to support students' collaborative programming learning. A quasi-experimental study was conducted in a junior high school programming course in Taiwan to assess the effects on students. The impacts of MR-CPS from both individual and collaborative perspectives were investigated. Students' learning achievement and computational thinking tendencies were examined from an individual perspective. From a collaborative perspective, group self-efficacy and group metacognition were investigated. Participants were divided into MR-CPS ($n = 115$) and No-MR-CPS ($n = 107$). The MR-CPS group used the collaborative programming approach with metacognitive regulation mechanisms as the experimental group. In contrast, the No-MR-CPS group used the collaborative programming approach without metacognitive regulation mechanisms as the control group. The

results show that the MR-CPS group statistically significantly outperformed the No-MR-CPS group in learning achievements. It was also found that the MR-CPS group had statistically significantly better computational thinking tendency, collective efficacy and metacognitive planning and evaluation skills than the No-MR-CPS group. This finding suggests that the MR-CPS has the potential to improve students' learning achievements, computational thinking tendency, group metacognition and collective efficacy. The study results have implications for the design of collaborative programming systems consistent with metacognitive regulation.

KEYWORDS

collaborative learning, computational thinking, metacognition, metacognitive regulation, programming education

Practitioner notes

What is already known about this topic

- Collaborative programming has markedly improved students' programming learning performance and computational thinking. However, it is a complex process and is not easy to access.
- Individual knowledge and metacognition skills are essential and challenging factors that could affect collaborative learning.

What this paper adds

- A collaborative programming approach based on metacognitive regulation was proposed to enhance students' academic performance during the learning process.
- With the proposed approach, develop a collaborative programming system based on metacognitive regulation to guide students to focus on metacognitive regulation during the collaborative programming process.
- The study results have implied that the Metacognitive Regulation-based Collaborative Programming System (MR-CPS) effectively improves students' learning achievements, computational thinking tendency, group metacognition and collective efficacy.

Implications for practice and/or policy

- The Metacognitive Regulation-based Collaborative Programming System (MR-CPS) is a potential approach to implementing collaborative programming activities.
- The relationship between the shift in students' metacognitive-regulated learning behaviours and their computational thinking development during collaborative programming activities deserves in-depth study.

INTRODUCTION

Researchers and educators have devoted significant attention to computational thinking (CT) because it is a twenty-first-century higher order thinking skill (Binkley et al., 2012) that is an integral part of modern research and problem solving in the STEM field of work (Wing, 2006). Programming, as a primary means of developing CT (Arfé et al., 2019; Román-González et al., 2017; Wing, 2006), is considered an excellent way to create computational work and demonstrate computational thinking skills (Grover & Pea, 2013). Researchers argue that learning programming allows students to think and act using skills related to computational thinking (eg, problem solving, decomposition, pattern abstraction, identification, testing and evaluation ... etc) (Buitrago Flórez et al., 2017; Polat & Yilmaz, 2022; Shute et al., 2017), thus enhancing students' computational thinking.

In addition, researchers believe that collaborative programming is conducive to developing students' computational thinking (Denner et al., 2014; Wu et al., 2019). Collaborative programming provides opportunities for students to communicate and share ideas, allows students to see different approaches to problem solving and helps students generate better ideas, thereby reducing deficiencies in programming solutions and improving computational thinking knowledge and skills (Wei et al., 2021). In collaborative learning, learners discover gaps between each other and gain new insights from their partners' experiences and knowledge through interactions with group members, such as communicating, sharing, arguing and resolving conflicts, leading to knowledge construction (Altintas et al., 2014; Cai & Gu, 2019).

Although collaborative programming has many benefits for students' development (Shadie et al., 2014; Tsompanoudi et al., 2016; Zheng et al., 2022), some challenges remain. Due to some students' inadequate metacognitive knowledge and abilities (Cho & Jonassen, 2002), students have difficulty using effective strategies when faced with complex topics. They cannot monitor and reflect on the learning process (Mayer, 1998), which results in poor collaborative learning and CT performance, such as patching and guessing (Wu et al., 2019). Furthermore, students struggle to build a shared understanding with group members when negotiating discussions (Wu et al., 2019), leading to process loss (Demir & Seferoglu, 2021). Therefore, there is a need for metacognitive guidance when students are engaged in collaborative learning.

Previous research has shown that providing students with effective support for metacognitive regulation mechanisms (eg, self-planning, self-monitoring and self-reflection) in a collaborative learning environment can stimulate more metacognitive behaviours and enhance learning achievements and collective efficacy (Zheng et al., 2022; Zhong et al., 2016). As beginners in programming languages, junior high school students may lack the necessary metacognitive skills to regulate collaborative learning effectively (Hadwin et al., 2018). Therefore, giving students appropriate guidance and encouraging them to pay attention to the learning process is vital to make more accurate cognitive judgements during collaborative programming.

Therefore, this study developed a collaborative programming approach and a learning system based on metacognitive regulation. The approach improved the learning effectiveness of collaborative programming by guiding students to set goals, monitor the collaborative learning process and self-evaluate. A quasi-experimental study was conducted to assess the system's effectiveness in promoting students' learning achievements, computational thinking tendency, group metacognition awareness and collective efficacy. The following four research questions were examined:

- **RQ1** – Can the MR-CPS significantly improve students' learning achievements compared to the No-MR-CPS?

- **RQ2** – Can the MR-CPS significantly improve students' computational thinking tendency compared to the No-MR-CPS?
- **RQ3** – Can the MR-CPS significantly improve students' group metacognition compared to the No-MR-CPS?
- **RQ4** – Can the MR-CPS significantly improve students' collective efficacy compared to the No-MR-CPS?

LITERATURE REVIEW

Computational thinking and programming

Computational thinking (CT) is generally considered a cognitive skill using computational methods and tools to solve problems (Lye & Koh, 2014; Wing, 2006). CT has become a fundamental literacy and skill everyone must possess today (Shute et al., 2017). As a result, researchers and educators are increasingly interested in incorporating CT into K-12 education (Lai & Wong, 2022), hoping that future citizens can apply computational thinking knowledge and skills in a wide range of contexts (Boom et al., 2022).

Programming education is the most commonly used method for developing CT (Lye & Koh, 2014) because coding allows for thinking and acting using students' CT skills (eg, problem solving, decomposition, pattern abstraction and recognition, debugging and evaluation) (Buitrago Flórez et al., 2017; Polat & Yilmaz, 2022; Shute et al., 2017). Previous research has also validated the impact of programming on promoting K-12 students' CT skills (Grover & Pea, 2013). For example, Sáez-López et al. (2016) taught a programming course to secondary school students, who showed improvements in computational thinking and computational practices.

However, most learners still consider programming complex and challenging (Akinola, 2016; Cheng et al., 2013; Kucuk & Sisman, 2017). Programming as a logic and problem-solving activity requires complex thinking skills that students often find difficult to perform independently (Shadiev et al., 2014). Therefore, finding an appropriate pedagogical approach to help students learn to program better is challenging for computational thinking education.

Collaborative programming

Collaborative programming is an approach to teaching programming (Shadiev et al., 2014). This pedagogical approach stems from cognitive and social constructivist learning theories (Kalaian & Kasim, 2015). The theory suggests that in collaborative programming, a group of students negotiates and explores programming problems, which can trigger cognitive conflict and knowledge transfer, ultimately leading to knowledge construction. Collaborative programming is particularly useful in developing CT skills and knowledge of computational programming (Denner et al., 2014; Iskrenovic-Momcilovic, 2019). For example, Zhong et al. (2016) found that a collaborative programming learning approach helped enrich primary school students' programming knowledge and develop their CT and creative thinking skills. Collaborative programming learning environments also reduce students' negative emotions and enhance their self-efficacy (Garcia, 2021; Wei et al., 2021). In addition, researchers often use technology tools to support collaborative programming (Rum & Ismail, 2017; Zheng et al., 2022). Due to the support of technology for learning (Gallagher & Gallagher, 2013), collaborative programming effectively improves students' cognitive performance and motivation (Shadiev et al., 2014; Tsompanoudi et al., 2016).

However, collaborative learning is not always successful when students have opportunities to work together (Kreijns et al., 2003). Similarly, Soller (2001) reported that simply dividing students into different groups does not necessarily lead to effective social interaction. Successful collaborative learning depends on the efforts of everyone. In other words, learners need to have a plan for their learning and be able to monitor, coordinate and evaluate their learning activities during collaborative programming (Jeong & Hmelo-Silver, 2016). Nevertheless, researchers have found that individuals often fail to adequately control and monitor their cognitive learning activities during collaborative processes (Barron, 2003; Hadwin & Oshige, 2011). Therefore, it is necessary to use appropriate regulation strategies to enhance the effectiveness of collaborative learning by guiding learners to plan appropriately, monitor the process and reflect on the effects of collaborative programming.

Metacognitive regulation

Metacognition refers to “cognition of cognition”, which is the knowledge about the regulation of cognitive activities during learning (Flavell, 1979). Metacognition coordinates learning by planning, monitoring and evaluating cognitive processes (Schraw & Moshman, 1995), and it is a cognitive-level regulatory activity (De Backer et al., 2022; Flavell, 1979). Self-regulation is the awareness and behaviour students adopt during learning, involving metacognitive, motivational and behavioural components (Zimmerman, 1995). Although metacognition is not the same concept as self-regulation, several researchers have argued that metacognition plays a crucial role in self-regulation (White, 1998; Zepeda & Nokes-Malach, 2023). To be more precise, metacognition is made up of two key elements. One is metacognitive knowledge, which is awareness of our thinking processes and what we know. The other is metacognitive regulation, which refers to the activities we engage in to facilitate learning (Sandi-Urena et al., 2010; Tanner, 2012).

The importance of metacognitive regulation for individual and collaborative learning performance is well supported by relevant research. For example, Mohd Rum and Ismail (2017) found that learners with metacognitive scaffolding outperformed learners without metacognitive scaffolding. Moreover, metacognitive scaffolds during collaborative learning can improve students' learning achievement and programming skills (Zheng et al., 2022). Kwon et al. (2013) provided learners with a metacognitive scaffold in collaborative learning that can promote collaborative interaction and improve students' personal and collaborative performance by helping students complete planning, monitoring and reflection. Computational thinking is a process that involves formulating problems and solutions (Wing, 2006) and metacognition plays a central role in problem solving (Zepeda & Nokes-Malach, 2023). To solve a problem, not only do the various stages of problem solving need to be executed, but they also need to be consciously tracked, regulated and controlled. One study found that students' computational thinking is positively related to metacognitive regulation ability (Şen, 2022). Through metacognitive prompts, students' computational thinking can be significantly improved (Chen et al., 2021).

Although collaborative programming allows students to acquire programming knowledge and develop higher order thinking, research has shown that students need metacognitive regulation during collaborative learning activities to monitor their cognitive situation and make more rational decisions (Koriat et al., 2006). Past research on metacognitive regulation has focused on individual learning (Järvelä et al., 2015). The same attention needs to be paid to students' metacognitive regulation skills in a collaborative learning environment (Jeong & Hmelo-Silver, 2016) because in relatively open collaborative learning environments, students frequently struggle to apply sufficient metacognitive regulation skills spontaneously and therefore struggle to achieve satisfactory learning outcomes (Manlove et al., 2007). In

other words, students may not use metacognitive skills to regulate their cognitive learning activities during collaborative processes (Barron, 2003). Moreover, the more the complex programming activities are, the more metacognitive skills are required (Nurulain Mohd Rum & Zolkepli, 2018).

The process of collaborative learning generally involves three levels of metacognitive regulation, namely, metacognitive regulation of individual student cognition, co-regulation and socially shared regulation (De Backer et al., 2022; Iiskala et al., 2015). However, the main subject of metacognition is still the individual, regardless of whether the individual works in a team or independently (Kim et al., 2013). This study developed a collaborative programming approach and learning system based on individual metacognitive regulation to guide learners to activate metacognition in collaborative programming, thereby improving collaborative learning outcomes.

THE DEVELOPMENT OF A METACOGNITIVE REGULATION-BASED COLLABORATIVE PROGRAMMING SYSTEM

Flavell (1979) argued that the critical skills of metacognitive regulation are planning, monitoring and evaluation, which occurs before starting the task, during the execution and after the job respectively. Based on Flavell's (1979) work, this study proposed a metacognitive regulation-based collaborative programming approach to enhance the learning effect of collaborative programming and students' computational thinking and metacognition. As shown in Figure 1, the approach consists of seven stages. Among them, 'Set learning goals,' 'self-monitoring' and 'self-evaluation' are the three critical steps of metacognitive regulation and supervision that can occur at any stage of learning. Metacognitive regulation strategies are

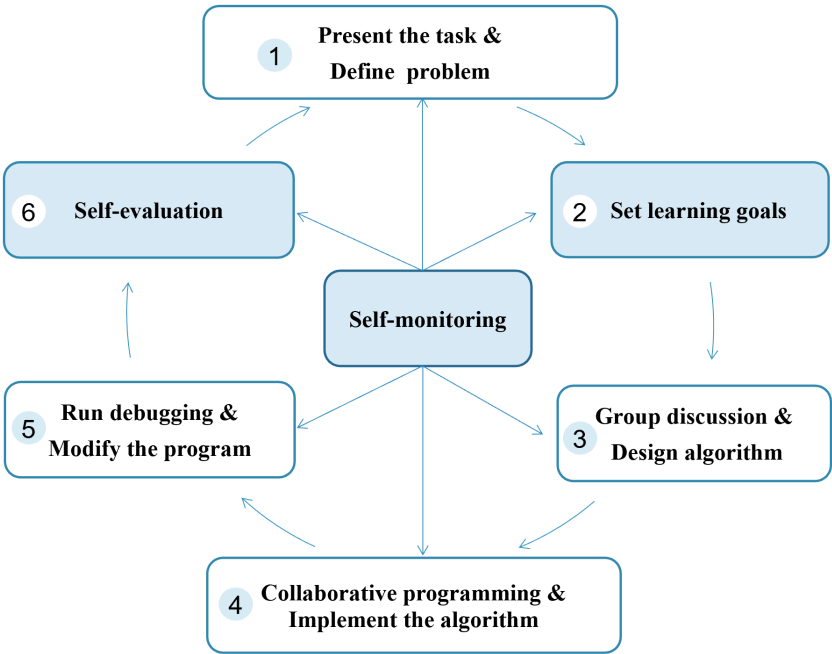


FIGURE 1 Metacognitive regulation-based collaborative programming approach.

highlighted in blue in Figure 1. Furthermore, metacognitive regulation is an iterative loop process. That is, the evaluation results may influence the goal setting for the next task.

At the beginning of the learning activity, the teacher presents the learning task and defines the problem by creating a real problem situation. Then, the students clarify the problems to be solved and set their individual learning goals from learning achievement, learning strategy selection and collaborative activities to participate in. After that, they engage in group discussions and design algorithms for the solution, which could be represented by drawing a programming flowchart. After the students have come up with a solution to the problem, they implement the solution's algorithm through collaborative programming and solve problems together after running debugging and modifying their programs continuously. Finally, students are guided to self-evaluate and reflect on their collaborative process and performance to adjust to the next programming learning activity. Throughout the collaborative process, students are guided to self-monitor their learning. Throughout the learning process, teachers need to observe students' collaborative status, prompt them to complete each collaborative task and provide guidance on any problems they encounter. In addition, after students submit their programming assignments, teachers need to assess them promptly and give students feedback and learning advice.

Based on the metacognitive regulation-based collaborative programming approach, the study developed a collaborative programming system to guide students in metacognitive regulation during collaborative programming. Figure 2 shows the system architecture. It mainly consists of a collaborative learning module and a metacognitive regulation module. The collaborative learning module supports students' collaborative activities, and the metacognitive regulation module provides a metacognitive scaffold to guide students in regulating and monitoring their cognitive activities during the problem-solving process. In addition, the collaborative programming environment includes an identity verification module, a task assignment module, a task guidance module, a metacognitive scaffolding editing module, a teacher evaluation module, a learning performance visualization module and databases. Teachers can turn off or on the metacognitive regulation module. In other words, the system has two modes. One is computer-supported collaborative programming with the metacognitive regulation module. The other is the conventional computer-supported collaborative programming mode, which does not allow the use of the metacognitive regulation module.

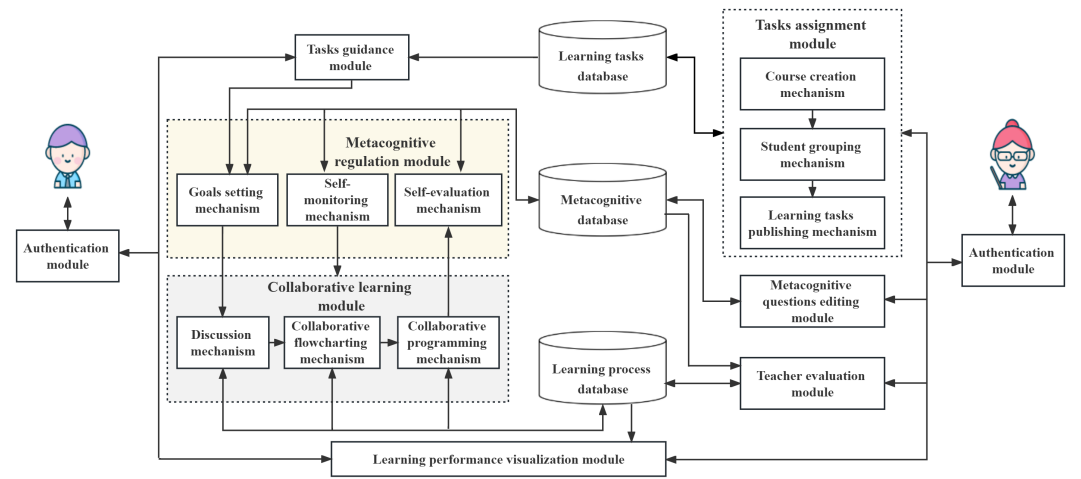


FIGURE 2 System architecture of collaborative programming based on metacognitive regulation.

First, the instructor logs into the learning system through the identity verification module. Before students can work on the activities, teachers need to create courses, publish learning tasks and group students through the learning task assignment module. In addition, the learning system has preset some metacognitive regulation questions. Teachers can also set other metacognitive regulation questions in the metacognitive regulation editing module according to the requirements of the learning content. After the instructor completes the task assignment and metacognitive scaffolding setup, students can log into the system and begin learning. Students are guided through the task guidance module to select the programming task they need to complete and learn about the requirements, the knowledge points involved and the deadline for submitting the assignment. The interface for collaborative programming tasks is shown in Figure 3.

After students understand the learning task, the system guides them to the metacognitive regulation module to set learning goals from individual and collaborative perspectives. The system guides students to use metacognition from an individual perspective through two questions: “How many points do I want to get?” and “What learning strategies do I plan to use?”. The system also guides students to use metacognition from a collaborative perspective through the question, “What tasks do I want to be actively involved in during collaborative programming?”. Each question will provide some options for students to choose from, guiding them to set their learning goals. Students must finish the goal setting before collaborating in a learning activity. The learning goal-setting interface is shown in Figure 4. The system records the student's options and compares them when the student completes the self-evaluation phase. When evaluating students, teachers will give learning suggestions considering the goals they set and the performance they achieved.

Next, students move to the collaborative learning module. The collaborative learning module consists of three learning mechanisms: the discussion mechanism, the collaborative flowcharting mechanism and the collaborative programming mechanism. The system allows students to collaborate on flowcharting and programming to complete a task simultaneously. During the problem-solving process of completing a task, students can discuss



FIGURE 3 The task interface.

the problems encountered at any time and solve the problem together. Different from other collaborative learning platforms/systems, the system provides a dual space where the problem solving and the social interaction spaces are closely integrated. Some researchers suggested constructing a dual-interaction space, where one is a workspace, and the other is an online chat room (Perit Çakır et al., 2009). In this way, students can effectively organize the social and cognitive processes involved in problem-solving activities. The interface of the collaborative programming and discussing interface is shown in Figure 5.

Researchers argued that if students do not know which strategies are available and when to use them, students will not be able to implement them in groups (Lyons et al., 2021). Therefore, the monitoring mechanism of this system differs from other meta-cognitive moderation systems in that monitoring is done by giving appropriate question

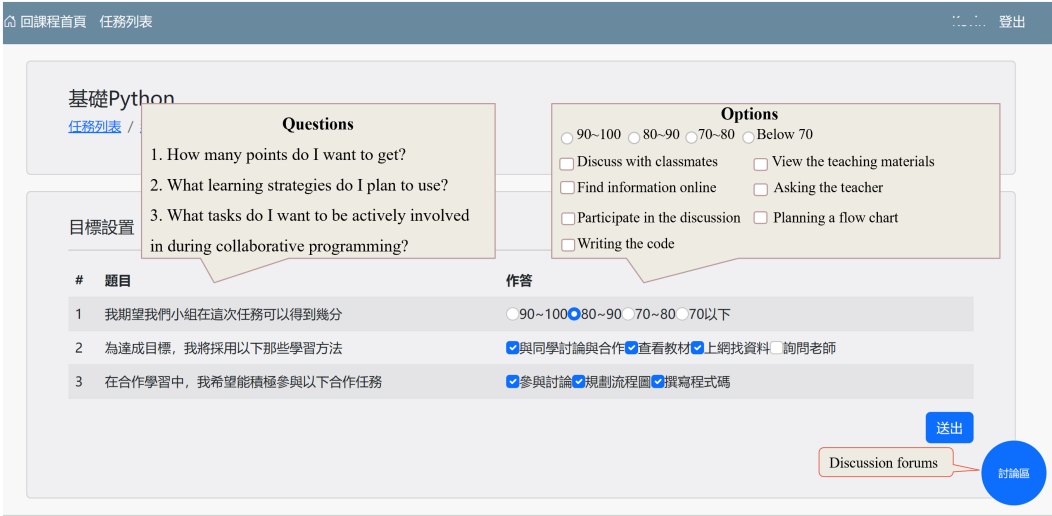


FIGURE 4 The learning goal-setting interface.

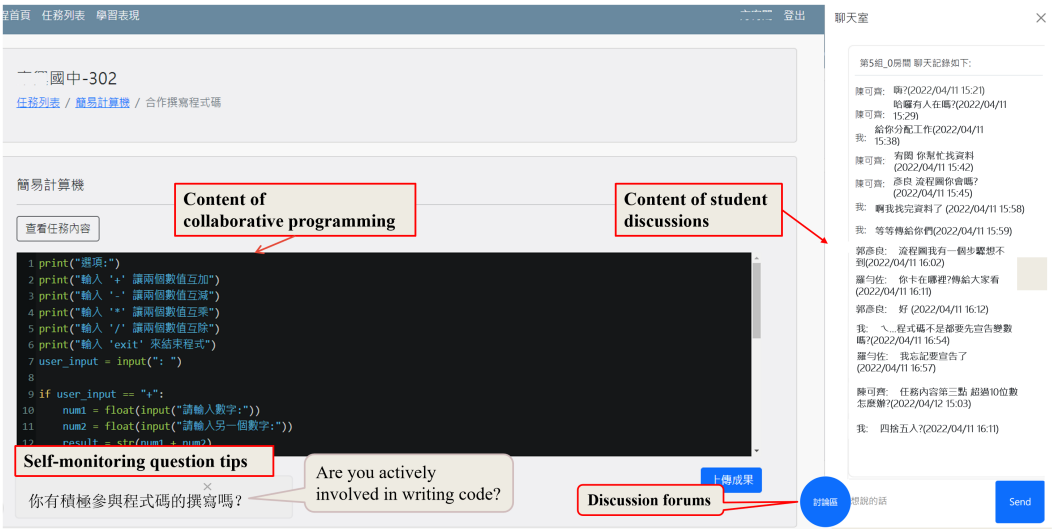


FIGURE 5 Collaborative programming and discussing interface.

prompts during the progress of collaborative programming, guiding students to monitor their collaborative learning process. For example, during group discussions, the system prompts, “Did you participate actively in the group discussion and present your ideas?” When writing code collaboratively, the system prompts, “Did you participate in writing code collaboratively?” Students need not answer these questions. The prompts only serve as gentle reminders to help students monitor their collaborative learning process and guide them towards their goals. The interface of self-monitoring is shown in the lower-left corner of Figure 5.

After students complete their collaborative programming tasks and submit their assignments, the instructor evaluates them through the evaluation module. Students can view their group's performance through the learning performance visualization module. Finally, the system guides students to self-evaluate and reflect on their learning performance. As shown in Figure 6, each student completes four self-evaluation questions: How many points do I think I can score for this collaborative performance? What strategies or methods did I use in this collaborative programming activity? What learning activities did I engage in during this collaborative programming learning? Based on the above reflections, how can I improve my next task?

METHODOLOGY

Participants

The participants were 222 students in their third year of junior high school in Hsinchu City, Taiwan, with an average age of 15. Coming from eight different classes, they all had enrolled in a junior high school IT course for 2.5 years and were all familiar with the basic operation of computers. There were no significant differences in the students' pretest ($F=0.85$, $p=0.36>0.05$), pre-computational thinking tendency ($F=0.28$, $p=0.36>0.05$), pre-group metacognition ($F=6.25$, $p=0.31>0.05$) and pre-collective efficacy ($F=3.90$, $p=0.06>0.05$). These results implied that students were at similar levels. Four classes of 115 students (60 boys, 55 girls) were assigned randomly as the MR-CPS group, and another four classes

The screenshot displays a self-evaluation interface. At the top, there are navigation links: '回課程首頁' and '任務列表'. Below this, a header bar shows '國中-302' and '任務列表 / 簡易計算機 / 自我反思'. The main content area is titled '自我反思' and contains four numbered questions in a light yellow box:

1. How many points do I think our group can score for this collaborative task?
2. What strategies did I use in this collaborative programming task?
3. What learning activities did I engage in during this collaborative programming learning?
4. Based on the above reflections, how can I improve in my next task?

Below the questions is a table with two columns: '# 題目' and '作答'. The table contains four rows of student responses. A red box highlights the first two rows, and another red box highlights the last two rows. A callout box labeled 'One student's choices and responses' points to the second row of the table.

# 題目	作答
1 我覺得我們小組在這次任務可以得到幾分	90~100
2 在本次學習任務中,我採用了以下那些學習方法	與同學合作討論,查看教材
3 在本次學習任務中,我積極參與了以下哪些合作任務	參與討論,規劃流程圖,撰寫程式碼
4 依據上述反思,你覺得下個任務可以如何改進	我覺得自己對程式代碼仍然有些不熟,希望下一次的程式代碼任務可以收穫滿滿!還有,希望可以參與討論!

At the bottom right of the table, there is a blue button labeled '編輯問答題'.

FIGURE 6 Self-evaluation interface.

of 107 students (52 boys, 55 girls) were the No-MR-CPS group. During the experiment, all students were randomly assigned to groups of 3–5 students to participate in collaborative programming activities. All groups were taught by the same instructor with over 5 years of experience teaching programming.

Participants volunteered to take part in this study. The study provided participants and their parents with an informed consent form explaining the experiment's details. The data collected were kept confidential, and coding was used to protect the names of the subjects from being made public. The experiment lasted from April to June 2022.

Instrument

The measurement instruments used in this study included the pretest, posttest and questionnaires on computational thinking tendency, group metacognition and collective efficacy. The purpose of the pretest was to assess learners' knowledge before the experiment, while the purpose of the posttest was to evaluate their learning achievement at the end of the experiment session. The pretest and posttest were designed by two junior schoolteachers with over 10 years of programming teaching experience through discussion. The teaching content was determined by negotiation between the experimenters and the instructor. Both tests consisted of 10 multiple-choice questions and were scored out of 100. Their main content is the basic programming concepts (data types, arithmetic operators, expressions, input and output functions, sequential, branch and loop structure, etc).

The computational thinking tendency questionnaire was adapted from a measure developed by Hwang et al. (2020) to examine the impact of the MR-CPS on students' computational thinking tendencies. It consists of six items (eg, I can usually develop a step-by-step procedure for solving complex problems), and Cronbach's α of the adapted questionnaire was 0.84.

The group metacognition questionnaire was employed to clarify learners' knowledge of cognition and metacognitive skills during collaborative learning. The group metacognition questionnaire used in this study was adapted from the measure developed by Biasutti and Frate (2018). Previous studies have used it to assess group metacognition for K-12 students (Socratous & Ioannou, 2022). The questionnaire was constructed of 18 items based on four dimensions: knowledge of cognition, planning skill, monitoring skill and evaluation skill (eg, I know how to use the material.). The total Cronbach's α of the adapted questionnaire was 0.89, and Cronbach's α of the four dimensions were 0.77, 0.80, 0.72 and 0.71 respectively.

The collective efficacy questionnaire was adapted from a measure developed by Wang and Lin (2007) to investigate the impact of the MR-CPS on students' collective efficacy. It consists of eight items (eg, I believe that our group can achieve a superior outcome for this learning task), and Cronbach's α was 0.91.

All questionnaires were administered using a 5-point scale, where 1–5 represented, in order, *complete disagreement*, *disagreement*, *neutrality*, *agreement* and *complete agreement*.

Experimental procedure

As shown in Figure 7, the experimental course lasted 7 weeks, with 45 minutes of class time per week. In the first week, students in both classes completed a pretest with a prequestionnaire. Subsequently, the instructor introduced the students to the learning approach and the

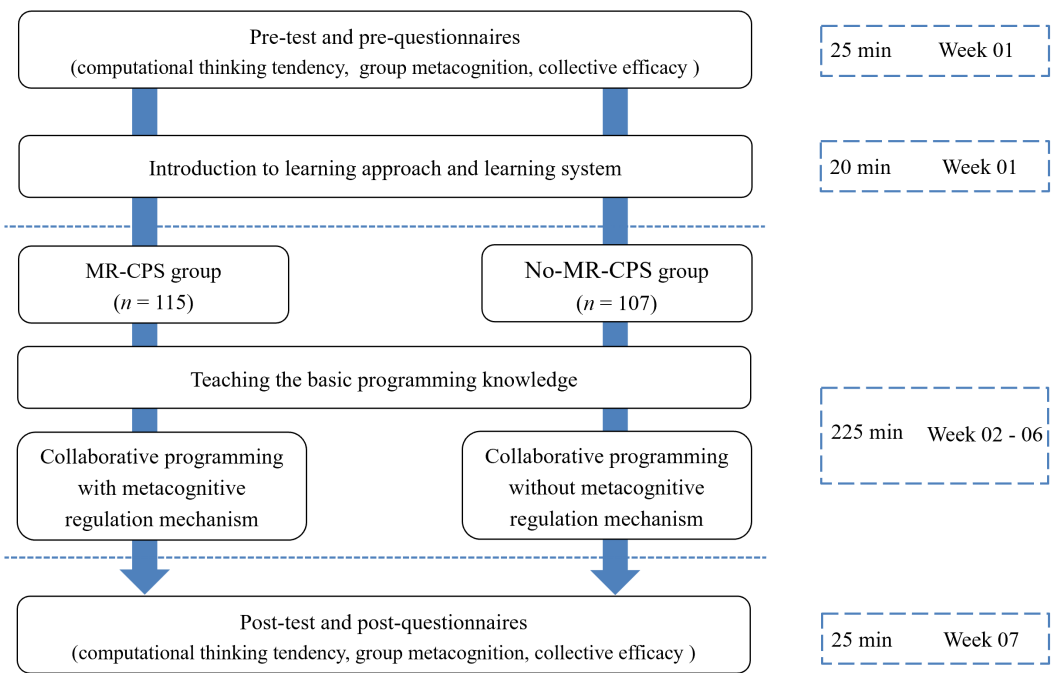


FIGURE 7 Experimental procedure diagram.

collaborative programming system to ensure that both classes understood the collaborative programming system for this study.

Formal programming instruction took place from week 2 to week 6. In each class, the instructor taught the appropriate basic programming knowledge. Then, the MR-CPS group performed collaborative programming activities under the metacognitive regulation-based collaborative programming system, while the No-MR-CPS group learned with the conventional collaborative programming system. Each group needed to complete Task 1 (design a simple calculator) within 2 weeks and Task 2 (design a complex calculator) within 3 weeks. Students in the MR-CPS group were required to complete metacognitive conditioning activities in the same amount of learning time. A detailed description of the tasks and schedule is given in Table 1. In the 7th week, all students completed a posttest and postquestionnaires.

Data analysis

SPSS 26 statistical software was used to analyse the data in this study. The quantitative data were assessed for conformity to a normal distribution by looking at Q–Q plots of the data distribution and the Shapiro–Wilk test. The results showed that all experimental data were normally distributed. In addition, Levene's test was used to test the homogeneity of variance of the quantitative experimental data. The results showed that the hypothesis of homogeneity of variance was valid for the learning achievements ($F=0.099$, $p=0.753$), computational thinking tendency ($F=0.243$, $p=0.624$), group metacognition ($F=2.408$, $p=0.122$) and group efficacy ($F=0.147$, $p=0.702$), demonstrating that it was possible to use analysis of variance.

To explore the effect of the proposed learning method on the students, an independent t -test was used to analyse the pretest and the prequestionnaire before the experiment. The results showed no significant difference between the two groups, as shown in Table 2. In other words, they could be considered parallel groups. After the experiment, to exclude the

effects of the pretest and prequestionnaire results, one-way ANCOVA was used to analyse the students' learning achievements, computational thinking tendencies, group metacognition and collective efficacy. The independent variable was the learning approach, while pretest and prequestionnaire ratings were the covariates, and posttest scores and postquestionnaire ratings were the dependent variable.

RESULTS

Learning achievements

The one-way ANCOVA results for posttest learning achievements are shown in Table 3. The adjusted mean scores for the experimental and control classes were $M=54.005$ and $M=47.752$. According to the results, the MR-CPS group had significantly higher learning achievements than the No-MR-CPS group ($F(1, 219)=4.344, p=0.038$) with a smaller effect size ($\eta^2=0.019$) (Cohen, 1988). The results imply that the MR-CPS is better at improving learning achievements in collaborative programming.

TABLE 1 Tasks and schedule for collaborative programming activities.

Tasks	Major program concepts	Description	Schedule
Task 1 Design a simple calculator	1. Data Types 2. Arithmetic operators 3. Expressions 4. Input and output functions 5. Sequential structure	Students shall collaborate to complete an algorithm flowchart and a simple calculator program, which should allow users to enter two positive numbers and perform addition and subtraction operations in Python.	Weeks 2–3
Task 2 Design a complex calculator	1. Data Types 2. Arithmetic operators 3. Expressions 4. Input and output functions 5. Sequential structure 6. Branch and Loop structure	Students shall work together to complete an algorithm flowchart and a program of a complex calculator, which can calculate the sum, the subtraction, the multiplication, the division, the average of two numbers and the square of one number in Python.	Weeks 4–6

TABLE 2 The *t*-test analysis of the pretest and postquestionnaire.

Variable	Groups	<i>N</i>	Mean	<i>SD</i>	<i>t</i>
Pre-LA	MR-CPS	115	42.783	21.665	0.923
	No-MR-CPS	107	45.327	19.197	
Pre-CTt	MR-CPS	115	3.103	0.790	−0.525
	No-MR-CPS	107	3.050	0.709	
Pre-GM	MR-CPS	115	3.735	0.532	−2.501
	No-MR-CPS	107	3.567	0.464	
Pre-CE	MR-CPS	115	3.662	0.788	−1.974
	No-MR-CPS	107	3.468	0.661	

Abbreviations: CE, collective efficacy; CTt, computational thinking tendency; GM, group metacognition; LA, learning achievements.

Computational thinking tendency

To evaluate the effects of the different learning approaches on computational thinking tendencies, the results of the one-way ANCOVA are shown in Table 4. The MR-CPS group had significantly higher computational thinking tendencies than the No-MR-CPS group ($F(1, 219)=12.404, p=0.001$), with a small-to-medium effect size ($\eta^2=0.054<0.059$) (Cohen, 1988). The adjusted mean scores for the MR-CPS and No-MR-CPS groups were $M=3.494$ and $M=3.175$ respectively. Based on these results, we can conclude that learners using the MR-CPS had better computational thinking tendency.

Group metacognition

In terms of group metacognition, the results of the one-way ANOVA are presented in Table 5. The MR-CPS group had significantly higher group metacognition than the No-MR-CPS group ($F(1, 219)=5.421, p=0.021$), with a small effect size ($\eta^2=0.024$) (Cohen, 1988). The adjusted mean for the MR-CPS and No-MR-CPS groups was $M=3.851$ and $M=3.697$ respectively. That is, the MR-CPS can better promote students' group metacognition development.

Table 6 further shows the students' levels in each dimension of group metacognition. There were significant differences between students in the MR-CPS and No-MR-CPS groups for planning and evaluation awareness. The planning awareness of the MR-CPS group was significantly better than those of the No-MR-CPS groups ($F(1, 219)=5.845, p=0.016$) with a smaller effect size ($\eta^2=0.026$) (Cohen, 1988). The adjusted mean scores for the MR-CPS

TABLE 3 The one-way ANCOVA analysis of learning achievements.

Groups	<i>N</i>	Mean	<i>SD</i>	Adjusted mean	<i>SE</i>	<i>F</i>	<i>p</i>	η^2
MR-CPS group	115	53.478	24.281	54.005	2.081	4.344*	0.038	0.019
No-MR-CPS group	107	48.318	23.532	47.752	2.154			

* $p<0.05$.

TABLE 4 The one-way ANCOVA analysis of computational thinking tendency.

Groups	<i>N</i>	Mean	<i>SD</i>	Adjusted mean	<i>SE</i>	<i>F</i>	<i>p</i>	η^2
MR-CPS group	115	3.506	0.778	3.494	0.063	12.404**	0.001	0.054
No-MR-CPS group	107	3.162	0.743	3.175	0.065			

** $p<0.01$.

TABLE 5 The one-way ANCOVA analysis of group metacognition.

Groups	<i>N</i>	Mean	<i>SD</i>	Adjusted mean	<i>SE</i>	<i>F</i>	<i>p</i>	η^2
MR-CPS group	115	3.891	0.581	3.851	0.046	5.421*	0.021	0.024
No-MR-CPS group	107	3.653	0.504	3.697	0.047			

* $p<0.05$.

TABLE 6 The one-way ANCOVA analysis of group metacognition in each dimension.

Dimensions	Groups	N	Mean	SD	Adjusted mean	SE	F	p	η^2
Planning	MR-CPS group	115	3.853	0.698	3.827	0.057	5.845*	0.016	0.026
	No-MR-CPS group	107	3.598	0.593	3.627	0.060			
Evaluation	MR-CPS group	115	3.919	0.669	3.908	0.055	6.917**	0.009	0.031
	No-MR-CPS group	107	3.687	0.576	3.699	0.057			
Knowledge of cognition	MR-CPS group	115	3.864	0.670	3.799	0.053	1.915	0.168	
	No-MR-CPS group	107	3.622	0.645	3.692	0.055			
Monitoring	MR-CPS group	115	3.928	0.635	3.899	0.060	3.709	0.550	
	No-MR-CPS group	107	3.706	0.659	3.737	0.058			

* $p < 0.05$; ** $p < 0.01$.

and No-MR-CPS groups were $M=3.827$ and $M=3.627$ respectively. In addition, the evaluation awareness of the MR-CPS group was significantly better than those of the No-MR-CPS group ($F(1, 219)=6.917, p=0.009$), with a smaller effect size ($\eta^2=0.031$) (Cohen, 1988). The adjusted mean scores for the MR-CPS and No-MR-CPS groups were $M=3.908$ and $M=3.699$ respectively. However, there were no significant differences between the two groups in terms of knowledge of cognition ($F(1, 219)=1.915, p=0.168$) or statistically significant differences in terms of monitoring awareness ($F(1,219)=3.709, p=0.550$). In other words, the MR-CPS was effective in terms of improving students' planning and evaluation awareness.

Collective efficacy

Table 7 shows the one-way ANCOVA results for collective efficacy. Students in the MR-CPS group had significantly higher collective efficacy than the No-MR-CPS group ($F(1, 219)=12.694, p<0.001$) with a small-to-medium effect size ($\eta^2=0.055<0.059$) (Cohen, 1988). The adjusted mean scores for the MR-CPS and No-MR-CPS groups were $M=3.852$ and $M=3.481$ respectively. These results suggest that the MR-CPS better enhances students' collective efficacy.

DISCUSSION AND CONCLUSIONS

This study developed a metacognitive-regulated collaborative programming approach and a programming learning system based on this approach. The results showed that MR-CPS could improve students' individual learning achievements and computational thinking tendency and also had a role in promoting students' group metacognition and collective efficacy. Therefore, the MR-CPS can provide students with metacognitive guidance to help them perform better on collaborative programming tasks and achieve better learning outcomes.

Regarding learning achievements, students in the MR-CPS group performed statistically significantly better than those in the No-MR-CPS group. This result is consistent with previous research findings (Theobald, 2021). It found that metacognition is a primary characteristic of high academic achievers (Wang et al., 1993). This result is similar to previous research findings that providing learners with self-planning, self-monitoring and self-evaluation mechanisms can significantly enhance their learning achievements (Lai & Hwang, 2016). Some studies have shown that students struggle to set goals and formulate plans during collaborative learning activities (Hadwin et al., 2018). The MR-CPS used metacognitive-regulated question scaffolding to guide learners in developing a learning plan in terms of learning objectives, learning methods and the collaborative tasks they planned to engage in, which helped students to have a clear plan for both learning objectives and collaborative assignments, and also facilitated the monitoring of the collaborative process and learning reflection (Hadwin et al., 2018), thus helping to improve their achievement (Çakıroğlu &

TABLE 7 The one-way ANCOVA analysis of collective efficacy.

Groups	N	Mean	SD	Adjusted mean	SE	F	p	η^2
MR-CPS group	115	3.872	0.784	3.852	0.072	12.694***	<0.001	0.055
No-MR-CPS group	107	3.460	0.779	3.481	0.075			

*** $p<0.001$.

Öztürk, 2017). Moreover, monitoring helps students become aware of various strategies that can be used and modified promptly to achieve learning goals (Zimmerman & Schunk, 2011). Research has shown that monitoring provides the basis for effective allocation of study time and effort, better control of attention and improved performance (Roebbers, 2017). Self-evaluation after completing a task leads to reflection on the achievement of learning objectives, the learning strategies used and the collaborative tasks in which they participated, which helps students to modify or refine their learning strategies to improve subsequent learning (Zimmerman, 2008). In terms of computational thinking tendency, students in the MR-CPS group performed statistically significantly better than those in the No-MR-CPS group. The result of this study implied that metacognitive regulation is a strategy that helps develop computational thinking. Computational thinking is constantly associated with problem solving (Kong et al., 2018). It has been found that self-reflection can improve students' ability to regulate and thus improve their problem-solving skills (Panaoura, 2012). In addition, developing a learning plan, monitoring the learning process, evaluating and reflecting can stimulate students' metacognition (Ouyang et al., 2022), which is significantly positively related to computational thinking (Şen, 2022) and therefore contributes to improving students' computational thinking (Liu et al., 2021; Liu & Liu, 2020).

Regarding group metacognition, students in the MR-CPS group expressed more confidence in planning and evaluating skills than those in the No-MR-CPS group. The possible reason is that the self-planning and self-evaluation mechanisms effectively activated the students' planning and evaluation awareness. This result aligns with previous research, which indicated that appropriate metacognitive scaffolding could stimulate students' metacognition (Ouyang et al., 2022). However, in terms of knowledge of cognition and monitoring, there were no statistically significant differences between the two groups. Sun et al. (2017) pointed out that external regulatory factors in regulative learning are susceptible to regulatory mechanisms (eg, seeking help, etc). In contrast, knowledge of cognition is an internal factor for learners. The reason may be that knowledge of cognition is the same between the two classes of students. In addition, because the self-monitoring mechanism in this system is presented in the form of a pop-up window, students may easily overlook it during the learning process. Therefore, the MR-CPS may not affect students' awareness and confidence in monitoring.

On the other hand, students who used the MR-CPS showed better collective efficacy than students who used the No-MR-CPS. Since collective efficacy refers to a group's shared beliefs about its perceived ability to perform a task to achieve a specified programming goal (Bandura et al., 1999), this finding implied that the self-planning and self-evaluation mechanisms in the MR-CPS significantly increased students' confidence and beliefs about the group's ability to complete the task. This finding is the same as previous studies (Zheng et al., 2022) because the MR-CPS helped students to control their learning process better and clarify their position in the group. The results of the study also indicated that students gained more relevant knowledge. As a result, students felt that they were more capable of completing learning tasks and were able to achieve higher scores through group work.

The experimental group was statistically significantly higher than the control group in terms of academic achievement, computational thinking tendencies, group metacognition and collective efficacy. The results suggest that the metacognitive-regulated collaborative programming approach has the potential to improve students' learning performance. However, the results also showed that the overall effect size was smaller. Therefore, the implications of the proposed approach for practical use should not be overstated. Experiments with longer intervention times could be conducted to verify further the effectiveness of the proposed metacognitive-regulated collaborative programming approach.

There are some limitations to this study. First, this study focused on the effects of the metacognitive regulation-based collaborative programming approach on junior high school

students. The findings cannot be extended to other school levels. Future research could focus on the effects of the approach on programming learning for elementary or college students. Second, in this study, no qualitative data were collected or analysed. Future studies could adopt a mixed research design. Third, the results of this study showed that students' self-monitoring awareness was not improved. The system needs to be further enhanced with the monitoring function and analysed with qualitative research data. Finally, different from previous studies that employed regulation only from an individual perspective, this study employed metacognitive regulation from individual and collaborative perspectives. However, the metacognition of the team as a whole, namely, socially shared regulation, is not involved in this study. Future research should consider designing socially shared metacognition that can stimulate group metacognition.

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CONFLICT OF INTEREST STATEMENT

All the authors have approved the manuscript and agree to submit it to *British Journal of Educational Technology*. There are no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

The analysed data can be provided upon request via e-mail to the corresponding author.

ETHICS STATEMENT

The study has been examined and advised by an academic ethics review committee.

ORCID

Wei Li  <https://orcid.org/0000-0002-4599-6433>

Cheng-Ye Liu  <https://orcid.org/0000-0003-0732-0288>

Judy C. R. Tseng  <https://orcid.org/0000-0002-3579-917X>

REFERENCES

- Akinola, S. O. (2016). Computer programming skill and gender difference: An empirical study. *American Journal of Scientific and Industrial Research*, 7(1), 1–9. <https://doi.org/10.5251/ajsir.2016.7.1.1.9>
- Altintas, T., Gunes, A., & Sayan, H. (2014). A peer-assisted learning experience in computer programming language learning and developing computer programming skills. *Innovations in Education and Teaching International*, 53(3), 329–337. <https://doi.org/10.1080/14703297.2014.993418>
- Arfé, B., Vardanega, T., Montuori, C., & Lavanga, M. (2019). Coding in primary grades boosts children's executive functions. *Frontiers in Psychology*, 10, 2713. <https://doi.org/10.3389/fpsyg.2019.02713>
- Bandura, A., Freeman, W. H., & Lightsey, R. (1999). *Self-efficacy: The exercise of control*.
- Barron, B. (2003). When smart groups fail. *The Journal of the Learning Sciences*, 12(3), 307–359. https://doi.org/10.1207/S15327809JLS1203_1
- Biasutti, M., & Frate, S. (2018). Group metacognition in online collaborative learning: Validity and reliability of the group metacognition scale (GMS). *Educational Technology Research and Development*, 66(6), 1321–1338. <https://doi.org/10.1007/s11423-018-9583-0>
- Binkley, M., Erstad, O., Herman, J., Raizen, S., Ripley, M., Miller-Ricci, M., & Rumble, M. (2012). Defining twenty-first century skills. In P. Griffin, & E. Care (Eds.), *Assessment and teaching of 21st century skills* (pp. 17–66). Springer. https://doi.org/10.1007/978-94-007-2324-5_2
- Boom, K. D., Bower, M., Siemon, J., & Arguel, A. (2022). Relationships between computational thinking and the quality of computer programs. *Education and Information Technologies*, 27(6), 8289–8310. <https://doi.org/10.1007/s10639-022-10921-z>

- Buitrago Flórez, F., Casallas, R., Hernández, M., Reyes, A., Restrepo, S., & Danies, G. (2017). Changing a generation's way of thinking: Teaching computational thinking through programming. *Review of Educational Research*, 87(4), 834–860. <https://doi.org/10.3102/0034654317710096>
- Cai, H., & Gu, X. (2019). Supporting collaborative learning using a diagram-based visible thinking tool based on cognitive load theory. *British Journal of Educational Technology*, 50(5), 2329–2345. <https://doi.org/10.1111/bjet.12818>
- Çakıroğlu, Ü., & Öztürk, M. (2017). Flipped classroom with problem based activities: Exploring self-regulated learning in a programming language course. *Educational Technology & Society*, 20(1), 337–349. <https://doi.org/10.2307/jeductechsoci.20.1.337>
- Chen, C., Liu, T., & Huang, K. (2021). Scaffolding vocational high school students' computational thinking with cognitive and metacognitive prompts in learning about programmable logic controllers. *Journal of Research on Technology in Education*, 55(3), 527–544. <https://doi.org/10.1080/15391523.2021.1983894>
- Cheng, C. C., Huang, P. L., & Huang, K. H. (2013). Cooperative learning in Lego robotics projects: Exploring the impacts of group formation on interaction and achievement. *Journal of Networks*, 8(7), 1529–1535. <https://doi.org/10.4304/jnw.8.7.1529-1535>
- Cho, K. L., & Jonassen, D. H. (2002). The effects of argumentation scaffolds on argumentation and problem solving. *Educational Technology Research and Development*, 50(3), 5–22.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences*. Routledge.
- De Backer, L., Van Keer, H., De Smedt, F., Merchie, E., & Valcke, M. (2022). Identifying regulation profiles during computer-supported collaborative learning and examining their relation with students' performance, motivation, and self-efficacy for learning. *Computers & Education*, 179, 104421. <https://doi.org/10.1016/j.compedu.2021.104421>
- Demir, Ö., & Seferoglu, S. S. (2021). A comparison of solo and pair programming in terms of flow experience, coding quality, and coding achievement. *Journal of Educational Computing Research*, 58(8), 1448–1466. <https://doi.org/10.1177/0735633120949788>
- Denner, J., Werner, L., Campe, S., & Ortiz, E. (2014). Pair programming: Under what conditions is it advantageous for middle school students? *Journal of Research on Technology in Education*, 46(3), 277–296. <https://doi.org/10.1080/15391523.2014.888272>
- Flavell, J. H. (1979). Metacognition and cognitive monitoring: A new area of cognitive-developmental inquiry. *American Psychologist*, 34(10), 906–911. <https://doi.org/10.1037/0003-066x.34.10.906>
- Gallagher, S. A., & Gallagher, J. J. (2013). Using problem-based learning to explore unseen academic potential. *Interdisciplinary Journal of Problem-Based Learning*, 7(1), 111–131. <https://doi.org/10.7771/1541-5015.1322>
- Garcia, M. B. (2021). Cooperative learning in computer programming: A quasi-experimental evaluation of jigsaw teaching strategy with novice programmers. *Education and Information Technologies*, 26(4), 4839–4856. <https://doi.org/10.1007/s10639-021-10502-6>
- Grover, S., & Pea, R. (2013). Computational thinking in K-12: A review of the state of the field. *Educational Researcher*, 42(1), 38–43. <https://doi.org/10.3102/0013189X12463051>
- Hadwin, A., & Oshige, M. (2011). Self-regulation, coregulation, and socially shared regulation: Exploring perspectives of social in self-regulated learning theory. *Teachers College Record*, 113(2), 240–264. <https://doi.org/10.1177/016146811111300204>
- Hadwin, A. F., Bakhtiar, A., & Miller, M. (2018). Challenges in online collaboration: Effects of scripting shared task perceptions. *International Journal of Computer-Supported Collaborative Learning*, 13(3), 301–329. <https://doi.org/10.1007/s11412-018-9279-9>
- Hwang, G. J., Li, K. C., & Lai, C. L. (2020). Trends and strategies for conducting effective STEM research and applications: A mobile and ubiquitous learning perspective. *International Journal of Mobile Learning and Organisation*, 14(2), 161–183. <https://doi.org/10.1504/ijmlo.2020.10022182>
- Iiskala, T., Volet, S., Lehtinen, E., & Vauras, M. (2015). Socially shared metacognitive regulation in asynchronous CSCL in science: Functions, evolution, and participation. *Frontline Learning Research*, 3, 78–111. <https://doi.org/10.14786/flr.v3i1.159>
- Iskrenovic-Momcilovic, O. (2019). Pair programming with scratch. *Education and Information Technologies*, 24(5), 2943–2952. <https://doi.org/10.1007/s10639-019-09905-3>
- Järvelä, S., Kirschner, P. A., Panadero, E., Malmberg, J., Phielix, C., Jaspers, J., Koivuniemi, M., & Jarvenoja, H. (2015). Enhancing socially shared regulation in collaborative learning groups: Designing for CSCL regulation tools. *Educational Technology Research and Development*, 63(1), 125–142. <https://doi.org/10.1007/s11423-014-9358-1>
- Jeong, H., & Hmelo-Silver, C. E. (2016). Seven affordances of computer-supported collaborative learning: How to support collaborative learning? How can technologies help? *Educational Psychologist*, 51(2), 247–265. <https://doi.org/10.1080/00461520.2016.1158654>

- Kalaian, S. A., & Kasim, R. M. (2015). Small-group vs. competitive learning in computer science classrooms: A meta-analytic review. In R. Queirós (Ed.), *Innovative teaching strategies and new learning paradigms in computer programming* (pp. 46–64). IGI Global. <https://doi.org/10.4018/978-1-4666-7304-5.ch003>
- Kim, Y. R., Park, M. S., Moore, T. J., & Varma, S. (2013). Multiple levels of metacognition and their elicitation through complex problem-solving tasks. *The Journal of Mathematical Behavior*, 32(3), 377–396. <https://doi.org/10.1016/j.jmathb.2013.04.002>
- Kong, S. C., Chiu, M. M., & Lai, M. (2018). A study of primary school students' interest, collaboration attitude, and programming empowerment in computational thinking education. *Computers & Education*, 127, 178–189. <https://doi.org/10.1016/j.compedu.2018.08.026>
- Koriat, A., Ma'ayan, H., & Nussinson, R. (2006). The intricate relationships between monitoring and control in metacognition: Lessons for the cause-and-effect relation between subjective experience and behavior. *Journal of Experimental Psychology: General*, 135(1), 36–69. <https://doi.org/10.1037/0096-3445.135.1.36>
- Kreijns, K., Kirschner, P. A., & Jochems, W. (2003). Identifying the pitfalls for social interaction in computer-supported collaborative learning environments: A review of the research. *Computers in Human Behavior*, 19(3), 335–353. [https://doi.org/10.1016/S0747-5632\(02\)00057-2](https://doi.org/10.1016/S0747-5632(02)00057-2)
- Kucuk, S., & Sisman, B. (2017). Behavioral patterns of elementary students and teachers in one-to-one robotics instruction. *Computers & Education*, 111, 31–43. <https://doi.org/10.1016/j.compedu.2017.04.002>
- Kwon, K., Hong, R.-Y., & Laffey, J. M. (2013). The educational impact of metacognitive group coordination in computer-supported collaborative learning. *Computers in Human Behavior*, 29(4), 1271–1281. <https://doi.org/10.1016/j.chb.2013.01.003>
- Lai, C. L., & Hwang, G. J. (2016). A self-regulated flipped classroom approach to improving students' learning performance in a mathematics course. *Computers & Education*, 100, 126–140. <https://doi.org/10.1016/j.compedu.2016.05.006>
- Lai, X. Y., & Wong, G. K. W. (2022). Collaborative versus individual problem solving in computational thinking through programming: A meta-analysis. *British Journal of Educational Technology*, 53(1), 150–170. <https://doi.org/10.1111/bjet.13157>
- Liu, S., & Liu, M. (2020). The impact of learner metacognition and goal orientation on problem-solving in a serious game environment. *Computers in Human Behavior*, 102, 151–165. <https://doi.org/10.1016/j.chb.2019.08.021>
- Liu, Y., Huang, T., & Sung, C. (2021). The determinants of impact of personal traits on computational thinking with programming instruction. *Interactive Learning Environments*, 1–15. <https://doi.org/10.1080/10494820.2021.1983610>
- Lye, S. Y., & Koh, J. H. L. (2014). Review on teaching and learning of computational thinking through programming: What is next for K-12? *Computers in Human Behavior*, 41, 51–61. <https://doi.org/10.1016/j.chb.2014.09.012>
- Lyons, K. M., Lobaczowski, N. G., Greene, J. A., Whitley, J., & McLaughlin, J. E. (2021). Using a design-based research approach to develop and study a web-based tool to support collaborative learning. *Computers & Education*, 161, 104064. <https://doi.org/10.1016/j.compedu.2020.104064>
- Manlove, S., Lazonder, A. W., & de Jong, T. (2007). Software scaffolds to promote regulation during scientific inquiry learning. *Metacognition and Learning*, 2(2–3), 141–155. <https://doi.org/10.1007/s11409-007-9012-y>
- Mayer, R. E. (1998). Cognitive, metacognitive, and motivational aspects of problem solving. *Instructional Science*, 26(1), 49–63. <https://doi.org/10.1023/A:1003088013286>
- Mohd Rum, S. N., & Ismail, M. A. (2017). Metacognitive support accelerates computer assisted learning for novice programmers. *Educational Technology & Society*, 20(3), 170–181.
- Nurulain Mohd Rum, S., & Zolkepli, M. (2018). Metacognitive strategies in teaching and learning computer programming. *International Journal of Engineering & Technology*, 7(4.38), 788–794. <https://doi.org/10.14419/ijet.v7i4.38.27546>
- Ouyang, F., Chen, S., Yang, Y., & Chen, Y. (2022). Examining the effects of three group-level metacognitive scaffoldings on in-service teachers' knowledge building. *Journal of Educational Computing Research*, 60(2), 352–379. <https://doi.org/10.1177/07356331211030847>
- Panaoura, A. (2012). Improving problem solving ability in mathematics by using a mathematical model: A computerized approach. *Computers in Human Behavior*, 28(6), 2291–2297. <https://doi.org/10.1016/j.chb.2012.06.036>
- Perit Çakır, M., Zemel, A., & Stahl, G. (2009). The joint organization of interaction within a multimodal CSCL medium. *International Journal of Computer-Supported Collaborative Learning*, 4(2), 115–149. <https://doi.org/10.1007/s11412-009-9061-0>
- Polat, E., & Yilmaz, R. M. (2022). Unplugged versus plugged-in: Examining basic programming achievement and computational thinking of 6th-grade students. *Education and Information Technologies*, 27(7), 9145–9179. <https://doi.org/10.1007/s10639-022-10992-y>
- Roebbers, C. M. (2017). Executive function and metacognition: Towards a unifying framework of cognitive self-regulation. *Developmental Review*, 45, 31–51. <https://doi.org/10.1016/j.dr.2017.04.001>
- Román-González, M., Pérez-González, J.-C., & Jiménez-Fernández, C. (2017). Which cognitive abilities underlie computational thinking? Criterion validity of the computational thinking test. *Computers in Human Behavior*, 72, 678–691. <https://doi.org/10.1016/j.chb.2016.08.047>

- Rum, S. N. M., & Ismail, M. A. (2017). Metacognitive support accelerates computer assisted learning for novice programmers. *Journal of Educational Technology & Society*, 20(3), 170–181. <http://www.jstor.org/stable/26196128>
- Sáez-López, J.-M., Román-González, M., & Vázquez-Cano, E. (2016). Visual programming languages integrated across the curriculum in elementary school: A two year case study using "scratch" in five schools. *Computers & Education*, 97, 129–141. <https://doi.org/10.1016/j.compedu.2016.03.003>
- Sandi-Urena, S., Cooper, M. M., & Stevens, R. H. (2010). Enhancement of metacognition use and awareness by means of a collaborative intervention. *International Journal of Science Education*, 33(3), 323–340. <https://doi.org/10.1080/09500690903452922>
- Schraw, G., & Moshman, D. (1995). Metacognitive theories. *Educational Psychology Review*, 7(4), 351–371. <https://doi.org/10.1007/bf02212307>
- Şen, Ş. (2022). Relations between preservice teachers' self-efficacy, computational thinking skills and metacognitive self-regulation. *European Journal of Psychology of Education*. <https://doi.org/10.1007/s10212-022-00651-8>
- Shadiev, R., Hwang, W. Y., Yeh, S. C., Yang, S. J. H., Wang, J. L., Han, L., & Hsu, G. L. (2014). Effects of unidirectional vs. reciprocal teaching strategies on web-based computer programming learning. *Journal of Educational Computing Research*, 50(1), 67–95. <https://doi.org/10.2190/EC.50.1.d>
- Shute, V. J., Sun, C., & Asbell-Clarke, J. (2017). Demystifying computational thinking. *Educational Research Review*, 22, 142–158. <https://doi.org/10.1016/j.edurev.2017.09.003>
- Socratous, C., & Ioannou, A. (2022). Evaluating the impact of the curriculum structure on group metacognition during collaborative problem-solving using educational robotics. *TechTrends*, 66(5), 771–783. <https://doi.org/10.1007/s11528-022-00738-5>
- Soller, A. (2001). Supporting social interaction in an intelligent collaborative learning system. *International Journal of Artificial Intelligence in Education*, 12, 40–62. <https://telearn.archives-ouvertes.fr/hal-00197321>
- Sun, J. C.-Y., Wu, Y.-T., & Lee, W.-I. (2017). The effect of the flipped classroom approach to OpenCourseWare instruction on students' self-regulation. *British Journal of Educational Technology*, 48(3), 713–729. <https://doi.org/10.1111/bjet.12444>
- Tanner, K. D. (2012). Promoting student metacognition. *CBE—Life Sciences Education*, 11(2), 113–120. <https://doi.org/10.1187/cbe.12-03-0033>
- Theobald, M. (2021). Self-regulated learning training programs enhance university students' academic performance, self-regulated learning strategies, and motivation: A meta-analysis. *Contemporary Educational Psychology*, 66, 101976. <https://doi.org/10.1016/j.cedpsych.2021.101976>
- Tsompanoudi, D., Satratzemi, M., & Xinogalos, S. (2016). Evaluating the effects of scripted distributed pair programming on student performance and participation. *IEEE Transactions on Education*, 59(1), 24–31. <https://doi.org/10.1109/te.2015.2419192>
- Wang, M. C., Haertel, G. D., & Walberg, H. J. (1993). Toward a knowledge base for school learning. *Review of Educational Research*, 63(3), 249–294. <https://doi.org/10.3102/00346543063003249>
- Wang, S. L., & Lin, S. S. J. (2007). The effects of group composition of self-efficacy and collective efficacy on computer-supported collaborative learning. *Computers in Human Behavior*, 23(5), 2256–2268. <https://doi.org/10.1016/j.chb.2006.03.005>
- Wei, X., Lin, L., Meng, N., Tan, W., & Kong, S. C. (2021). The effectiveness of partial pair programming on elementary school students' computational thinking skills and self-efficacy. *Computers & Education*, 160, 104023. <https://doi.org/10.1016/j.compedu.2020.104023>
- White, R. T. (1998). Decisions and problems in research on metacognition. In B. J. Fraser & K. G. Tobin (Eds.), *International handbook of science education* (pp. 1207–1213). Kluwer Academic Publishers.
- Wing, J. M. (2006). Computational thinking. *Communications of the ACM*, 49(3), 33–35. <https://doi.org/10.1145/1118178.1118215>
- Wu, B., Hu, Y. L., Ruis, A. R., & Wang, M. H. (2019). Analysing computational thinking in collaborative programming: A quantitative ethnography approach. *Journal of Computer Assisted Learning*, 35(3), 421–434. <https://doi.org/10.1111/jcal.12348>
- Zepeda, C. D., & Nokes-Malach, T. J. (2023). Assessing metacognitive regulation during problem solving: A comparison of three measures. *Journal of Intelligence*, 11, 16. <https://doi.org/10.3390/jintelligence11010016>
- Zheng, L., Zhen, Y., Niu, J., & Zhong, L. (2022). An exploratory study on fade-in versus fade-out scaffolding for novice programmers in online collaborative programming settings. *Journal of Computing in Higher Education*, 34(2), 489–516. <https://doi.org/10.1007/s12528-021-09307-w>
- Zhong, B. C., Wang, Q. Y., & Chen, J. (2016). The impact of social factors on pair programming in a primary school. *Computers in Human Behavior*, 64, 423–431. <https://doi.org/10.1016/j.chb.2016.07.017>
- Zimmerman, B. J. (1995). Self-regulation involves more than metacognition: A social cognitive perspective. *Educational Psychologist*, 30(4), 217–221. https://doi.org/10.1207/s15326985ep3004_8

- Zimmerman, B. J. (2008). Investigating self-regulation and motivation: Historical background, methodological developments, and future prospects. *American Educational Research Journal*, 45(1), 166–183. <https://doi.org/10.3102/0002831207312909>
- Zimmerman, B. J., & Schunk, D. H. (2011). *Handbook of self-regulation of learning and performance*. Routledge.

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